Graduate Project Report

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Summary

This project proposes a logistic regression model for disaster type classification and an ordinal logistic regression model for damage level classification based on satellite photos. Explanatory description analysis (EDA) is done through extracting the color, edge, and texture features from both the full image and the local blocks. For the first model, the K-fold cross-validation results indicate that the accuracy for the disaster type classification model reaches 0.9900. For the second model, we implemented a weighted logistic regression model for damage severity classification, which reached a 0.55 score. This report analyzes these limitations and proposes robust solutions to address this bias, including a cost-sensitive random forest and data augmentation. The link to the presentation video is https://youtu.be/l0zluHpp2U8.

1 Introduction

1.1 Background and Motivation

Rapid situational awareness after natural disasters is essential for emergency agencies to allocate resources and save lives. Because manual damage assessment from high-resolution satellite imagery is slow and labor-intensive, this project applies data science and computer vision to automate disaster-type classification and building-damage assessment from satellite images.

1.2 Survey of Related Work

The use of remote sensing data for disaster response has been a subject of extensive research. Early approaches primarily utilized handcrafted features, such as color histograms and texture descriptors, combined with traditional machine learning classifiers like Support Vector Machines (SVMs) or Random Forests[1]. These methods, while interpretable, often struggle with the high variability and noise inherent in satellite imagery.

Current state-of-the-art approaches largely leverage Deep Learning, specifically Convolutional Neural Networks (CNNs). The release of large-scale datasets, such as the xView2 Challenge dataset[2], has accelerated research in this domain. Recent methodologies, such as those by Gupta et al. (2019) and Weber and Kané (2020), typically employ semantic segmentation models (e.g., U-Net, ResNet-based architectures) to simultaneously localize buildings and classify damage on a pixel-by-pixel basis[2, 3]. These end-to-end deep learning pipelines have shown superior performance in handling complex environmental factors compared to feature-based methods.

1.3 Project Scope

Instead of jumping straight to complex, end-to-end semantic segmentation or black-box deep learning, this project takes a foundational, interpretable approach to disaster imagery by building a transparent preprocessing and feature-engineering pipeline. We explicitly isolate and analyze key visual cues—color, structure, and texture—and evaluate engineered features (e.g., Sobel edges for structural integrity and Local Binary Patterns for debris texture) during EDA to establish a strong baseline. The goal is to test whether simpler, computationally efficient features can reliably distinguish disaster types and damage patterns before moving to more resource-intensive neural network models.

2 Data Description and EDA

2.1 Dataset Summary

The dataset employed for this project is a subset of the xView2 Challenge Dataset, curated to simulate real-world crisis scenarios. The data is organized into three disaster types, allowing for the assessment of model generalization across different terrains and damage types:

- Midwest Flooding: Imagery characterized by water inundation and mud.
- Socal-fire: Imagery characterized by burn scars, ash, and smoke.
- Hurricane-matthew: Imagery characterized by structural wind damage and flooding.

Each data point consists of a high-resolution RGB satellite image (stored as a NumPy array) and a corresponding label file. The labels provide two target variables for modeling:

- Disaster Type: A categorical label identifying the event (Flood, Fire, or Hurricane).
- Damage Level: An ordinal scale ranging from 0 to 3.

2.2 Data Cleaning and Sampling

Before modeling, we perform data cleaning to check for any NaN values in the dataset. The results reveal that all 600 images across the three datasets are valid. However, there is a significant imbalance in the classes representing different damage levels. Classes 1 and 2 have very few samples, while Classes 0 and 3 have a larger number of samples. This imbalance could hinder our ability to accurately identify the rarer classes and differentiate them from the more abundant ones, ultimately affecting the overall accuracy and F1 score of the classification. To tackle this issue, we will conduct data augmentation to generate more samples for Class 2. Further details will be provided in the modeling section.

2.3 Exploratory Data Analysis

2.3.1 Analysis of Disaster Type

To capture visual differences across disaster types, we analyzed the distribution of median RGB intensities. Figure 1 shows clear separation in color signatures: **socal-fire** images have higher and more variable red-channel values, consistent with burned vegetation and exposed soil, while **midwest-flooding** images show elevated blue and green values, reflecting standing water and saturated terrain. **hurricane-matthew** falls between these extremes with moderately high green intensities typical of mixed vegetation and storm-affected areas. Overall, color features appear highly discriminative for classifying disaster type.

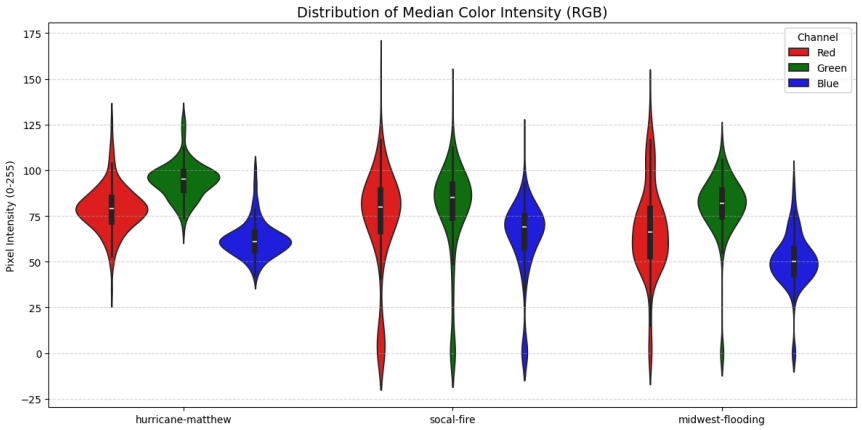


Figure 1. Distribution of Median Color Intensity by Disaster Type

To complement the color analysis, we evaluated structural characteristics using Sobel-based edge magnitude features. Figure 2 shows that *socal-fire* images tend to have slightly higher Sobel mean and standard deviation values, suggesting more fragmented, irregular textures caused by debris and fire-induced surface disruptions. *Midwest-flooding* shows lower and more compressed edge intensities, consistent with large smooth areas such as water bodies where edges are less prominent. *Hurricane-matthew* again falls between these extremes, reflecting the mixed presence of intact buildings and storm-generated damage. These outcomes indicate that edge-based features can further support disaster-type classification.

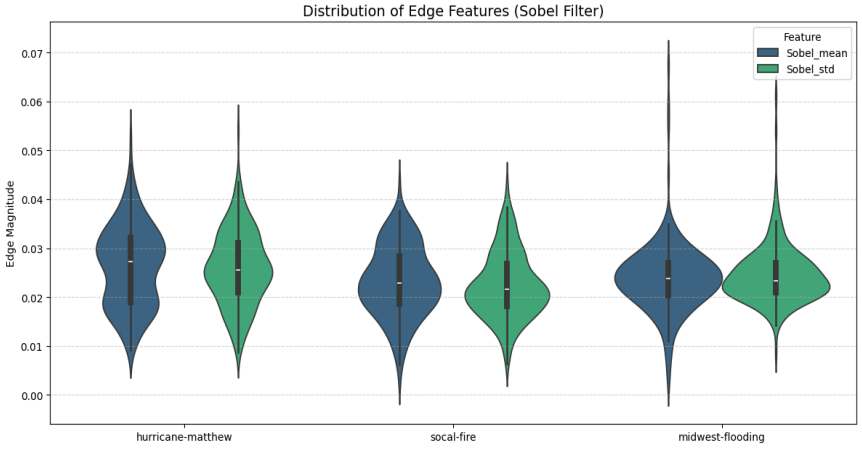


Figure 2. Distribution of Edge Features by Disaster Type

Texture differences were examined using Local Binary Pattern (LBP) quantiles. As shown in Figure 3, *socal-fire* images display lower and more compact lower-quartile LBP values, indicating more uniform surface textures following widespread burning. Conversely, *hurricane-matthew* and *midwest-flooding* images exhibit higher variability in LBP quantiles, reflecting the more complex and heterogeneous textures found in their environments. Collectively, the color, edge, and texture analyses confirm that each disaster scenario exhibits a distinct visual signature. This suggests that a feature-based model is well-suited for Task A, where the objective is to classify images into flooding or fire categories.

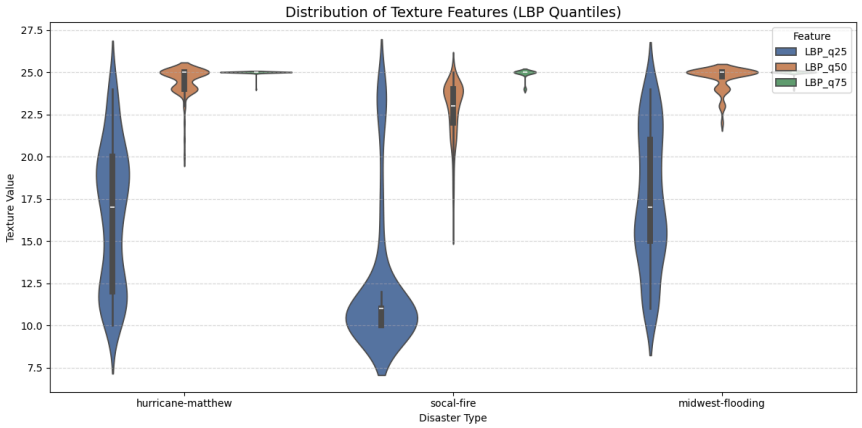


Figure 3. Distribution of Texture Features by Disaster Type

2.3.2 Analysis of Damage Level

Next, we tested whether our engineered features could distinguish the four damage levels (0–3) in the hurricane-matthew dataset. Figure 4 shows that Levels 0–1 have tight, stable median RGB distributions, consistent with intact roofs and uniform materials, while Levels 2–3 display wider ranges across all channels—likely reflecting exposed materials, debris, and uneven shading from severe damage. Overall, color helps separate minor from severe damage, but the overlap across levels suggests it's not sufficient on its own for precise classification.

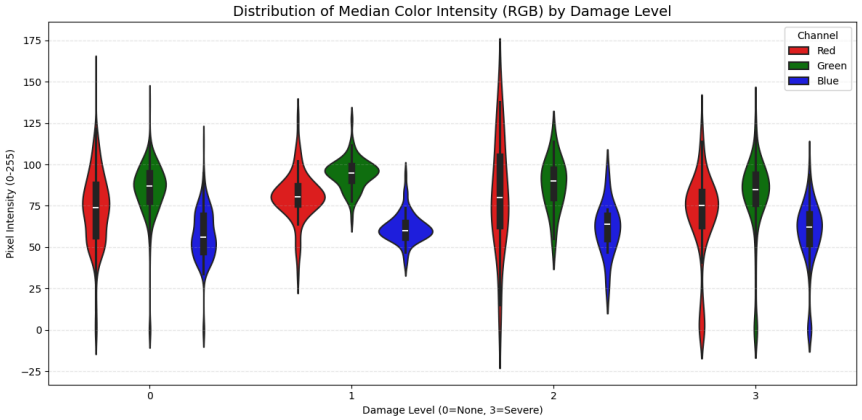


Figure 4. Distribution of Median Color Intensity by Damage Level

Edge-based features offer clearer separation, as illustrated in Figure 5. Buildings with no or light damage (Levels 0 and 1) exhibit lower and more tightly distributed Sobel values, indicating smooth and intact rooftops. Meanwhile, Levels 2 and 3 display both higher mean edge magnitudes and larger variance, reflecting structural fragmentation and collapsed debris that generate irregular edge patterns. These results suggest that edge features are particularly effective at identifying high levels of structural damage.

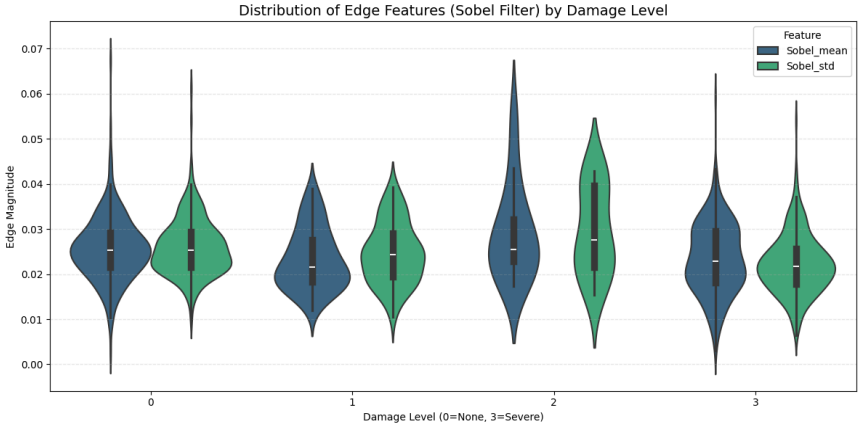


Figure 5. Distribution of Edge Features by Damage Level

The texture analysis using LBP quantiles reinforces this trend. Figure 6 shows that Levels 0 and 1 generally maintain lower LBP values characteristic of regular roof textures, while Levels 2 and 3 exhibit substantially higher variability, capturing the chaotic textures associated with exposed interiors, rubble, and roofing failures. These findings confirm that texture features provide strong secondary signals for identifying severe building damage.

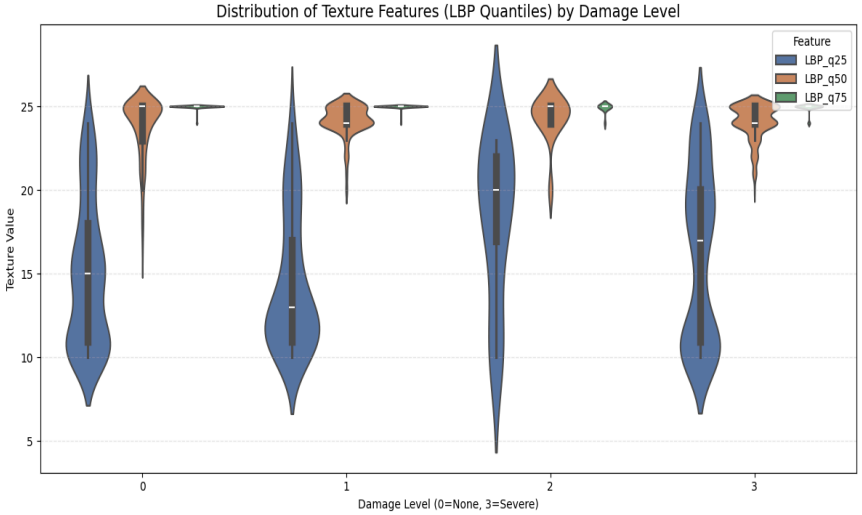


Figure 6. Distribution of Texture Features by Damage Level

3 Modeling

3.1 Disaster Type Classification

3.1.1 Model Construction

Task A is to build a classification model to distinguish images from “midwest-flooding” and “socal-fire” disasters. Since it is a binary classification problem, we build a logistic regression model. To reduce the model variance and overcome the overfitting problem, the model employs L2 regularization, which includes a hyperparameter λ. In this model, K-fold cross-validation is utilized to optimize this hyperparameter. Accuracy is used to evaluate the classification model.

3.1.2 Feature Engineering

(1) Color features: the mean values of each RGB channel for each image are calculated, based on the assumption that flooding and fire images are typically different in terms of color.
(2) Edge features: The Sobel algorithm is used to extract the edges from satellite images. Edge density is the average value of each pixel. In satellite imagery, terrain and topography are the main factors that influence edges and texture, and many of these features are localized. To enhance feature granularity, images are divided into 3×3 blocks, with average edge density calculated from each block. Edge feature contains 9 input variables.
(3) Texture features: The LBP algorithm extracts local texture features and outputs a new image. To aggregate the information of each image, the Q1, median, and Q3 of the pixel values for each processed image are calculated as the representation of texture information. Similarly, the image is divided into 2×2 blocks. Therefore, the texture feature contains 12 variables in total.

3.2 Damage Level Classification

3.2.1 Model Construction

Our modeling approach was iterative, driven by the need to handle both the complexity of the features and the severe class imbalance. We initially experimented with a Random Forest classifier to leverage its ability to capture non-linear relationships, but it failed to generalize effectively on our limited dataset. Given the inherent ranking of damage levels (where Class 0<1<2<3), we subsequently tested an Ordinal Logistic Regression model. However, this approach proved unsuitable because it relies on latent threshold constraints that assume a structured progression between categories; it could not account for the unequal visual distances between damage types (e.g., the visual "leap" from *no-damage* to *minor* differs significantly from *major* to *destroyed*). Consequently, we pivoted to a weighted multinomial logistic regression. This model offered the necessary flexibility by treating classes independently rather than imposing a rigid ordinal structure. To finally unlock the model's potential against the imbalance, we implemented a dual strategy: data augmentation (with geometric transformations by flipping horizontally the images) was applied strictly to the training set to expand the support for minority

classes, and cost-sensitive learning was enforced via aggressive class weights to explicitly penalize the misclassification of rare events.

3.2.2 Feature Engineering

To transform the high-dimensional raw pixel data into a robust input for our logistic regression, we engineered a compact 25-dimensional feature vector. We adopted a hybrid strategy to capture both the color shifts and the localized structural damage caused by the disaster. First, we extracted 9 Global Color features (RGB quartiles: Q1, Median, Q3) to characterize the overall palette of the scene, based on the rationale that flooding events induce a holistic color shift (e.g., towards muddy tones) independent of spatial location. Second, to detect physical destruction, we implemented a local zonal analysis by subdividing each image into a 2×2 grid (4 quadrants). Within each block, we computed the average edge density using sobel filters (4 features) to capture topography changes, and extracted texture statistics (Q1, Median, Q3 of local binary patterns) for each block (12 features). This spatial segmentation was crucial, allowing the model to identify localized debris fields or surface roughness that would be lost in a global average. We also used StandardScaler in our model to normalize our hybrid features, which was mandatory for this algorithm to converge given the diverse values of our features.

4 Inference and Analysis

4.1 Disaster Type Classification

The sizes of the training and validation sets are 8:2. To decide the L2 regularization hyperparameter c in the logistic regression model, we tried different values and implemented the model cross-validation for each choice. As shown in Figure 7, the final choice is: c=0.1, which has the highest validation accuracy, indicating its quality in fitting and generalization.

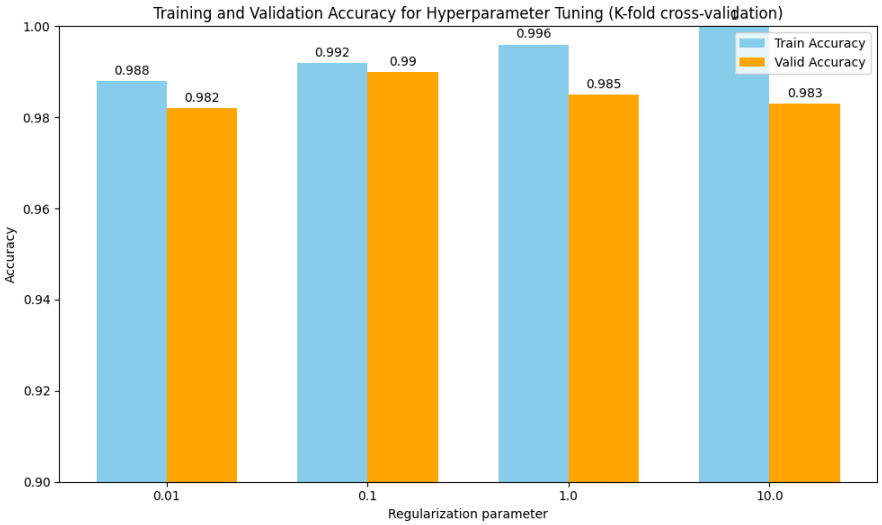


Figure 7. K-fold cross-validation for Disaster Type Classification

The classification accuracy of training and validation sets is shown in Table 2. Figure 8 shows the ROC and PR curves for the validation set. The average accuracy of the disaster type classification model reaches 0.9900 on the validation set, which proves that the logistic regression model performs well in the disaster type classification task with generalization ability.

Table 2. Classification accuracy of each fold

Dataset	K-fold cross-validation (K=5)				
	1	2	3	4	5
Training set	0.990625	0.99375	0.990625	0.990625	0.996875
Validation set	1.0000	0.9875	0.9875	1.0000	0.9750

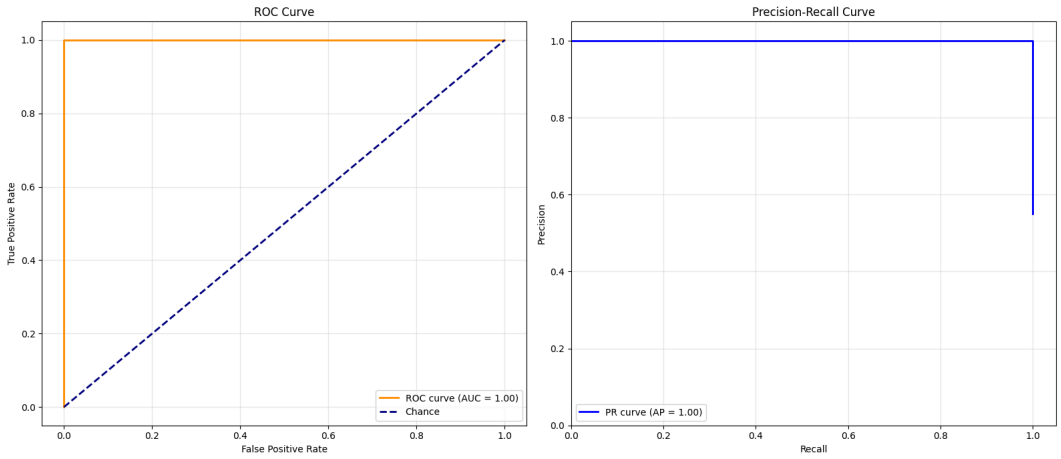


Figure 8. ROC curve and PR curve

4.2 Damage Level Classification

Table 3. Results of damage level classification

Class	Precision	Recall	F1-score	Support
0	0.38	0.50	0.43	6
1	0.67	0.43	0.52	14
2	0.00	0.00	0.00	1
3	0.92	0.63	0.75	19

To rigorously assess the model's generalization capabilities and mitigate the risk of overfitting -particularly given the use of synthetic data augmentation- we employed a stratified K-Fold

cross-validation. This approach ensured that each training fold maintained the representative class distribution, providing a stable estimate of the model's performance during the hyperparameter tuning phase. The cross-validation process yielded consistent F1-scores (averaging ~0.53), confirming the stability of our weighted logistic regression. Upon final evaluation on the completely unseen test set, the model achieved a Weighted F1-score of 0.55. While this metric highlights the persistent challenge of distinguishing the highly underrepresented intermediate classes (1 and 2), the solid performance on the extreme classes ('no-damage' and 'destroyed') validates the effectiveness of our hybrid feature extraction and cost-sensitive training strategy in identifying significant destruction.

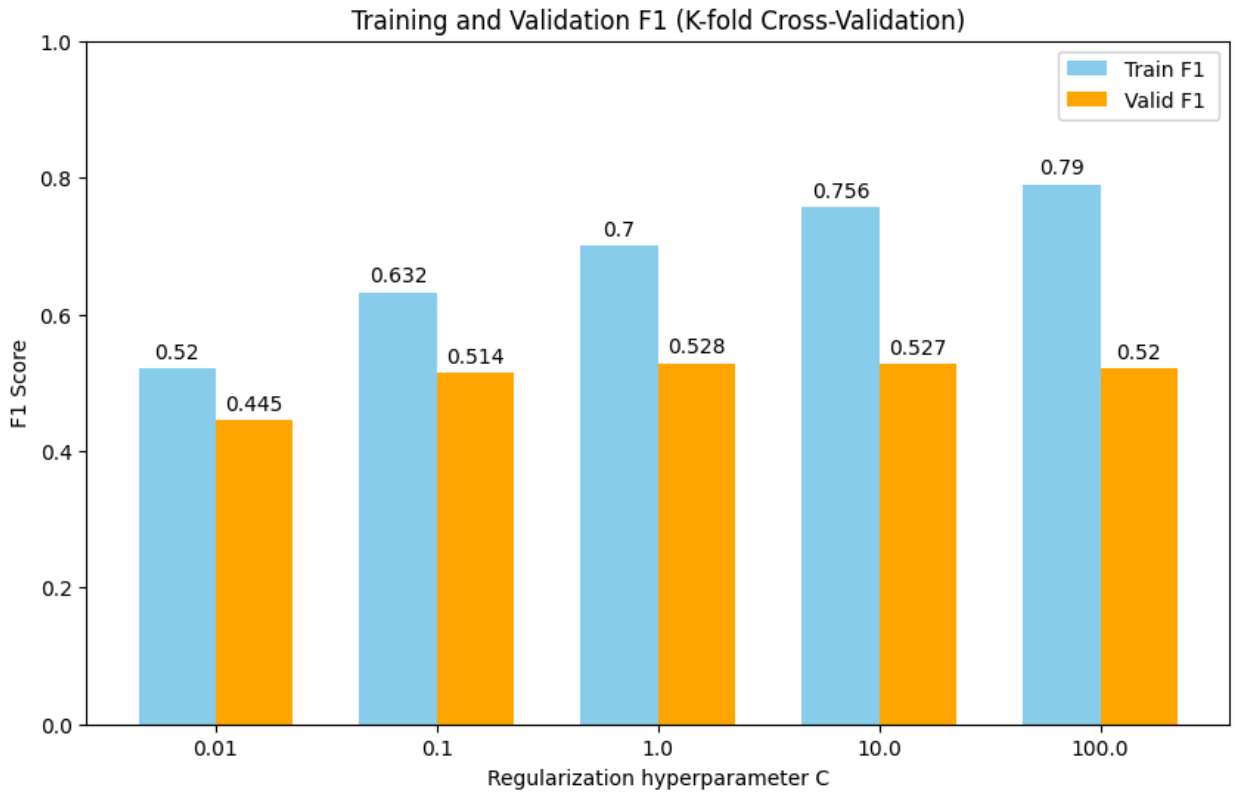


Figure 9. K-fold Cross-validation Damage Level Classification

5 Discussion

5.1 Strengths and Limitations

The proposed methods have several strengths. First, in terms of feature engineering, the image is divided into small blocks to extract local features, which enhances the spatial resolution. For the damage level classification model, we take into account the sequence of damage levels and utilize an ordinal logistic regression model. However, there are also some limitations to our approach. Unlike neural network techniques, such as convolutional neural networks, our method relies on

manual feature engineering, which may not effectively capture much of the local information within the images. By focusing on aggregated features of the entire image, we risk losing subtle details. Moreover, for the damage level classification, the models we employed are relatively simple in order to maintain interpretability. Although we have implemented data augmentation and employed a weighted loss function, the data imbalance issue remains. Further exploration of neural networks and other advanced methods might prove beneficial.

5.2 Societal impact analysis

There are social concerns about errors in the damage level prediction. When the damage level is wrongly predicted, we should be more concerned about underestimation. This is because if the predicted damage level is lower than the actual damage, then the disaster emergency response will be inadequate. This underestimation leads to more severe social consequences than overestimation. In contrast, the disaster type classification should have a higher accuracy, and fewer mistakes are permitted. For example, it is essential to clearly discriminate between fire and flooding. Otherwise, disaster response measures will be entirely misguided, resulting in immense waste and loss.

References:

[1] Nex, F., Remondino, F. UAV for 3D mapping applications: a review. Appl Geomat 6, 1–15 (2014).
[2] Gupta, Ritwik, et al. "xbd: A dataset for assessing building damage from satellite imagery." arXiv preprint arXiv:1911.09296 (2019).
[3] Weber, Ethan, and Hassan Kané. "Building disaster damage assessment in satellite imagery with multi-temporal fusion." arXiv preprint arXiv:2004.05525 (2020).